

How Correlation Breaks the Optimal Portfolio

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Abstract

The mean-variance framework prescribes an “optimal” portfolio, but its inputs must be estimated from data and therefore carry error. We ask how sensitive the resulting portfolio weights are to that error. We first derive the minimum-variance portfolio for two assets using single-variable calculus, then generalize to three or more assets via Lagrange multipliers. We apply the result to two datasets of daily log returns: a weakly-correlated portfolio (AAPL, KO, JPM) and a highly-correlated bank portfolio (JPM, BAC, WFC). Under identical perturbations to a single estimated variance, the maximum weight shift is roughly five times larger in the bank portfolio. We trace this contrast to the condition number of the covariance matrix, which rises from $\kappa \approx 2.50$ in the first case to $\kappa \approx 13.88$ in the second. A nearly proportional scaling between κ and weight sensitivity suggests fragility is controlled by how close the covariance matrix is to being singular.

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1 Introduction

After understanding more about how risks and returns work in our current stock market, I was curious to understand the relationship between risk and reward for multiple assets. As the saying goes, "Don't put all your eggs in one basket," I wanted to understand why this matters. Reading Markowitz (1952)'s paper on an "optimal" portfolio, I wanted to analyze real data and illustrate as well as extend on his findings. The concern that mean-variance optimization can amplify estimation error rather than dampen it which is often called "error maximization" was raised by Michaud (1989), and it gives us a reason to examine how sensitive the optimal weights really are.

2 The Minimum-Variance Problem

2.1 Setup

Consider a portfolio of assets with certain risks. For two assets with variances σ_1^2, σ_2^2 and covariance σ_{12} , an investor places weight w in the first asset and $1 - w$ in the second, so that the weights sum to one. We later generalize to n assets, where matrix notation becomes necessary.

2.2 The Two-Asset Case

The portfolio variance as a function of w is

$$\text{Var}(w) = w^2\sigma_1^2 + (1 - w)^2\sigma_2^2 + 2w(1 - w)\sigma_{12}. \quad (1)$$

Expanding and collecting terms by powers of w ,

$$\text{Var}(w) = (\sigma_1^2 + \sigma_2^2 - 2\sigma_{12})w^2 + (-2\sigma_2^2 + 2\sigma_{12})w + \sigma_2^2. \quad (2)$$

Differentiating with respect to w and setting the result to zero,

$$\frac{d \text{Var}(w)}{dw} = 2(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12})w + 2(\sigma_{12} - \sigma_2^2) = 0, \quad (3)$$

which yields the minimum-variance weight

$$w^* = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}. \quad (4)$$

Since the denominator equals $\text{Var}(r_1 - r_2)$, it is always positive whenever the two assets are not perfectly correlated. When $\sigma_{12} = 0$, the formula reduces to $w^* = \sigma_2^2/(\sigma_1^2 + \sigma_2^2)$, which places more weight on the lower-variance asset.

$$\frac{d^2 \text{Var}(w)}{dw^2} = 2(\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) = 2 \text{Var}(r_1 - r_2) > 0.$$

Furthermore, since the second derivative of $\text{Var}(w)$ is positive, we confirm that this critical point is a minimum.

3 The General Case

3.1 Why Scalars No Longer Suffice

The two-asset case was tractable because the single budget constraint $w_1 + w_2 = 1$ left exactly one free variable, reducing the problem to ordinary single-variable calculus. With more than two assets this is no longer true since after imposing the budget constraint, two or more weights remain free, and the bookkeeping of individual variances and pairwise covariances quickly becomes arduous. We therefore adopt vector and matrix notation, which expresses the problem compactly for any number of assets n and makes the structure of the solution transparent.

3.2 Setup

Consider a portfolio of n risky assets. Let

$$\mathbf{r} = (r_1, \dots, r_n)^\top \in \mathbb{R}^n$$

denote the random vector of asset returns, with expected-return vector

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{r}] \in \mathbb{R}^n$$

and covariance matrix

$$\boldsymbol{\Sigma} = \text{Cov}(\mathbf{r}) \in \mathbb{R}^{n \times n}, \quad \Sigma_{ij} = \text{Cov}(r_i, r_j).$$

We assume $\boldsymbol{\Sigma}$ is symmetric and positive definite, so that $\boldsymbol{\Sigma}^{-1}$ exists. A portfolio is described by a weight vector

$$\mathbf{w} = (w_1, \dots, w_n)^\top \in \mathbb{R}^n,$$

where w_i is the fraction of wealth allocated to asset i . Writing $\mathbf{1} = (1, \dots, 1)^\top$ for the vector of ones, the budget constraint is $\mathbf{w}^\top \mathbf{1} = 1$. The portfolio's expected return and variance are

$$\mathbb{E}[\mathbf{w}^\top \mathbf{r}] = \mathbf{w}^\top \boldsymbol{\mu}, \quad \text{Var}(\mathbf{w}^\top \mathbf{r}) = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}.$$

3.3 The Global Minimum-Variance Portfolio

We minimize the portfolio variance subject to the budget constraint, using a single Lagrange multiplier.

The global minimum-variance portfolio solves

$$\min_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^\top \mathbf{1} = 1. \tag{5}$$

Introducing a Lagrange multiplier λ , we form the Lagrangian

$$\mathcal{L}(\mathbf{w}, \lambda) = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{1} - 1). \tag{6}$$

Taking the gradient with respect to $\mathbf{w}$ and using the symmetry of Σ ,

$$\nabla_{\mathbf{w}}\mathcal{L} = (\Sigma + \Sigma^\top)\mathbf{w} - \lambda\mathbf{1} = 2\Sigma\mathbf{w} - \lambda\mathbf{1}. \quad (7)$$

Setting $\nabla_{\mathbf{w}}\mathcal{L} = \mathbf{0}$ and solving for $\mathbf{w}$,

$$2\Sigma\mathbf{w} = \lambda\mathbf{1} \implies \mathbf{w} = \frac{\lambda}{2}\Sigma^{-1}\mathbf{1}. \quad (8)$$

Substituting (8) into the budget constraint $\mathbf{w}^\top\mathbf{1} = 1$ gives

$$\frac{\lambda}{2}\mathbf{1}^\top\Sigma^{-1}\mathbf{1} = 1 \implies \lambda = \frac{2}{\mathbf{1}^\top\Sigma^{-1}\mathbf{1}}. \quad (9)$$

Back-substituting (9) into (8), the factors of 2 and λ cancel, yielding the global minimum-variance portfolio

$$\mathbf{w}^* = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top\Sigma^{-1}\mathbf{1}} \quad (10)$$

	AAPL	KO	JPM
Mean daily log return	0.001769	0.000626	0.000557
Daily variance	0.00019127	0.00009771	0.00018216

Table 1: Summary statistics for the three assets, computed from daily log returns over the period 2025-05-28 to 2026-05-27 ($T = 251$ observations).

$$\Sigma = \begin{pmatrix} 0.00019127 & 0.00001207 & 0.00004586 \\ 0.00001207 & 0.00009771 & -0.00000924 \\ 0.00004586 & -0.00000924 & 0.00018216 \end{pmatrix}$$

$$\mathbf{w}^* = (w_{\text{AAPL}}, w_{\text{KO}}, w_{\text{JPM}}) = (0.1773, 0.5478, 0.2749).$$

4 How Fragile Is the “Optimal” Portfolio?

4.1 A Weakly-Correlated Example

Having derived the global minimum-variance portfolio in closed form, we must now ask how much confidence we should place in the resulting weights. The inputs to the formula are not known exactly, for they are gathered from historical data and therefore carry estimation error. To find the portfolio’s sensitivity, we take the estimated covariance matrix as our base case and introduce a small, deliberate error into a single input. Then we recompute the optimal weights and measure how far they move. We perturb a variance rather than an expected return because the global minimum-variance portfolio depends only on the covariance matrix Σ and not on $\boldsymbol{\mu}$. Among the three assets we perturb AAPL’s variance, since AAPL has the largest estimated variance and the smallest base-case weight, making it

Perturbation	AAPL	KO	JPM	Max. shift (pp)
Base case	17.73%	54.78%	27.49%	—
+5% AAPL var	16.92%	55.22%	27.87%	0.82
+10% AAPL var	16.17%	55.61%	28.22%	1.56
+20% AAPL var	14.87%	56.30%	28.84%	2.87

Table 2: Global minimum-variance weights under increasing perturbations to AAPL’s estimated variance. “Max. shift” is the largest absolute change in any single weight, in percentage points, relative to the base case.

the asset the optimizer is working hardest to limit. We repeat the experiment at perturbation sizes of 5%, 10%, and 20% to see whether the response grows in a predictable way.

Based on these results, we find two important patterns: As AAPL’s estimated variance rises, the optimizer shifts weight away from AAPL and toward lower-variance assets, and the response is roughly linear in the size of the perturbation, with the largest weight change approximately doubling (from 0.82 to 2.87 percentage points) as the perturbation grows from 5% to 20%. Notably, even a 20% error in a single variance estimate moves the weights by under three percentage points. This relative stability reflects the fact that our three assets are weakly correlated, which keeps the covariance matrix well-conditioned; as we discuss in the next section, the sensitivity can become far more drastic when assets are highly correlated and the matrix approaches singularity.

4.2 A Highly-Correlated Comparison

To test whether the moderate sensitivity observed above is specific to our choice of weakly-correlated assets, we repeat the experiment on a second dataset chosen for the assets’ high interrelatedness. We use daily log returns for JPM, BAC, and WFC over the period 2025-05-30 to 2026-05-29 ($T = 251$ observations). Because all three banks are exposed to the same regulatory environment, interest-rate movements, and credit cycle, we expect their returns to be substantially more correlated than those in the previous example.

The estimated correlation matrix confirms this:

$$\text{Corr}(\mathbf{r}) = \begin{pmatrix} 1.00 & 0.74 & 0.66 \\ 0.74 & 1.00 & 0.80 \\ 0.66 & 0.80 & 1.00 \end{pmatrix},$$

with pairwise correlations ranging from 0.66 to 0.80, compared to the range of -0.07 to 0.25 in the first dataset. The resulting sample covariance matrix is

$$\Sigma = \begin{pmatrix} 0.00018250 & 0.00013354 & 0.00014935 \\ 0.00013354 & 0.00018077 & 0.00017856 \\ 0.00014935 & 0.00017856 & 0.00027838 \end{pmatrix},$$

and its base-case global minimum-variance weights are

$$\mathbf{w}^* = (w_{\text{JPM}}, w_{\text{BAC}}, w_{\text{WFC}}) = (0.5036, 0.5637, -0.0673).$$

Since WFC and BAC are highly correlated but not exactly the same, the math tells us to short WFC to allow more funds to go into JPM and BAC. The optimizer uses the small residual difference between two nearly-redundant assets by shorting one of the two, cancelling out most of the shared risk and leaving only a low-variance leftover. This itself is an early warning sign of the fragility we are about to examine. An extreme leveraged position resting on the small estimated difference between two near-twins becomes unstable under small input errors, as the perturbation experiment below makes concrete.

We now repeat the perturbation experiment, increasing JPM’s estimated variance by 5%, 10%, and 20% as before, and recomputing the GMV weights.

Perturbation	JPM	BAC	WFC	Max. shift (pp)
Base case	50.36%	56.37%	−6.73%	—
+5% JPM var	45.86%	60.07%	−5.93%	4.50
+10% JPM var	42.10%	63.17%	−5.27%	8.26
+20% JPM var	36.17%	68.06%	−4.22%	14.19

Table 3: Global minimum-variance weights for the bank portfolio under the same variance perturbations applied in Table 2.

The contrast with Table 2 is drastic. Under identical perturbations the maximum weight shift grew from 0.82% to 4.50% at +5%, from 1.56% to 8.26% at +10%, and from 2.87% to 14.19% at +20% which represents roughly a 5x increase in sensitivity. The condition number of the bank covariance matrix is $\kappa \approx 13.88$, compared to $\kappa \approx 2.50$ in the first study, an increase of a similar factor.

Based on these results, we find the same two patterns from the first study, but greatly amplified: as JPM’s estimated variance rises, the optimizer shifts weight away from JPM and toward BAC, and the response is again roughly linear in the size of the perturbation, with the largest weight change approximately tripling (from 4.50 to 14.19 percentage points) as the perturbation grows from 5% to 20%. The contrast with the first study is extreme. Under identical perturbations the maximum weight shift is roughly five times larger; a 20% error in a single variance estimate now moves a weight by over fourteen percentage points rather than under three. This instability reflects the fact that our three banks are strongly correlated, which pushes the covariance matrix toward singularity; the condition number of Σ rises from $\kappa \approx 2.50$ in the first study to $\kappa \approx 13.88$ here, an increase of a similar factor to that observed in the weights.

5 Why the Optimizer Is So Sensitive

We have derived the equation for the optimal weights, and in doing so found the important fact that the optimal weights depend on Σ^{-1} . The sensitivity we observed comes from the behavior of matrix inversion: under certain conditions, small changes in Σ can be greatly amplified when it is inverted.

Since taking the inverse of a matrix is analogous to $\frac{1}{x}$, we can use that functions characteristics to explain the large changes in the matrix. For example, for some large x , $\frac{1}{x}$ barely changes (e.g. $\frac{1}{100} \approx \frac{1}{99}$). However, for some $x \rightarrow 0$, $\frac{1}{x}$ drastically changes (e.g. $\frac{1}{.001} \not\approx \frac{1}{.0011}$). A matrix close to singularity behaves similarly to this function where small errors are magnified at high correlation (i.e. moving the matrix to singularity) and stable at low correlation.

When two assets are strongly correlated, their corresponding rows in Σ become nearly proportional, leaving the matrix close to being linearly dependent therefore close to singularity.

To make this precise, we use the condition number of the covariance matrix, which measures how strongly matrix inversion amplifies small errors. For a symmetric positive-definite matrix, it is defined as the ratio of the largest eigenvalue to the smallest,

$$\kappa(\Sigma) = \frac{\lambda_{\max}}{\lambda_{\min}},$$

and satisfies $\kappa \geq 1$, with values close to 1 indicating a well-conditioned (stable) matrix and large values indicating a near-singular (unstable) one. The eigenvalues of our estimated covariance matrix are approximately 0.000093, 0.000145, and 0.000233, giving

$$\kappa(\Sigma) = \frac{0.000233}{0.000093} \approx 2.50.$$

This value is very close to 1, confirming that our covariance matrix is well-conditioned. This is the direct cause of the stability we observed in Section 4: because κ is small, inverting Σ amplifies estimation errors only modestly, so even a 20% error in a single variance shifted the weights by under three percentage points. Had our assets been more strongly correlated, the smallest eigenvalue would have been driven toward zero, inflating κ and producing far larger swings in the optimal weights.

6 The Standard Remedy

One standard remedy is shrinkage (Ledoit and Wolf, 2004). One blends the sample covariance $\hat{\Sigma}$ with a stable target F via $\Sigma_{\text{shrunk}} = \alpha\hat{\Sigma} + (1 - \alpha)F$, where α controls the blend. Because the target is well-conditioned, this lifts the smallest eigenvalues away from zero, lowering κ toward 1 and stabilizing $\hat{\Sigma}^{-1}$.

7 Open Questions

While our two studies cover low and moderately high correlation, they represent only two points on a much wider spectrum. A more thorough study would sample many portfolios across a range of condition numbers and characterize the relationship between κ and weight sensitivity quantitatively, rather than relying on two case studies. Furthermore, we only used GMV and not the full frontier, removing μ from our calculations. μ is the hardest quantity to predict (Best and Grauer, 1991). Moreover, we used a single time window (one year of daily data). The covariance estimate itself depends on that choice, so a different window would give different numbers and potentially different stability.

Acknowledgments

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References

- Anthropic (2026). Claude (Opus 4.7). Large language model.
- Best, M. J. and Grauer, R. R. (1991). On the sensitivity of mean-variance-efficient portfolios to changes in asset means: Some analytical and computational results. *The Review of Financial Studies*, 4(2):315–342.
- Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411.
- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1):77–91.
- Michaud, R. O. (1989). The Markowitz optimization enigma: Is “optimized” optimal? *Financial Analysts Journal*, 45(1):31–42.